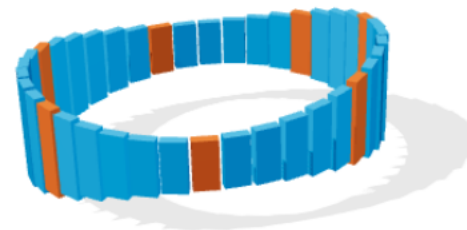


VLA- JEPA

Can watching
become doing?

Enhancing vision-language-action models with latent world prediction

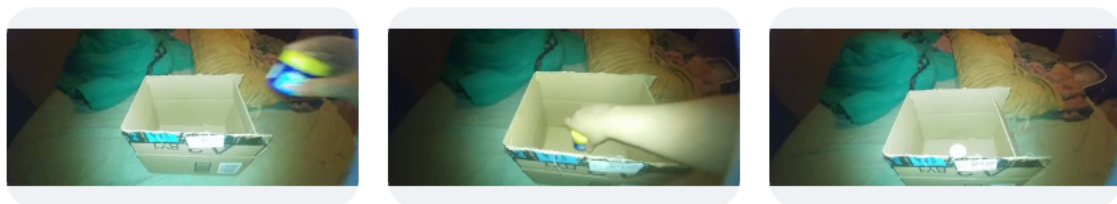


Transition → state → control

Video shows outcomes. Robot data supplies control.

WATCH

Human video



Before

During

After



“Put the jar into the box.”

Video + description; no robot control labels

ACT

Robot demonstration



“Place the green block inside the black bowl.”

Recorded pose commands · x/y/z normalized per axis



Feature prediction alone is not the whole distinction

LAPA

Pixels

Frame pair → discrete code → policy → robot adaptation

UniVLA

DINO features

Frame pair + language → discrete code → policy → action decoder

VLA-JEPA

V-JEPA 2 features

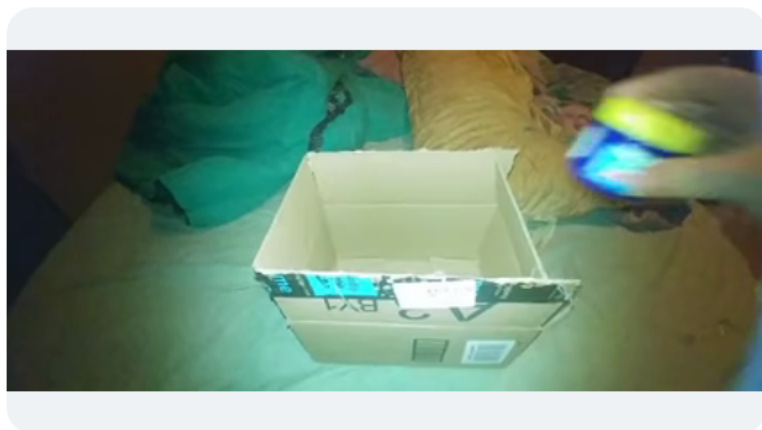
Current observation → continuous tokens → future-state prediction

VLA-JEPA directly trains the policy tokens through future-state alignment.

UniVLA already uses feature reconstruction.

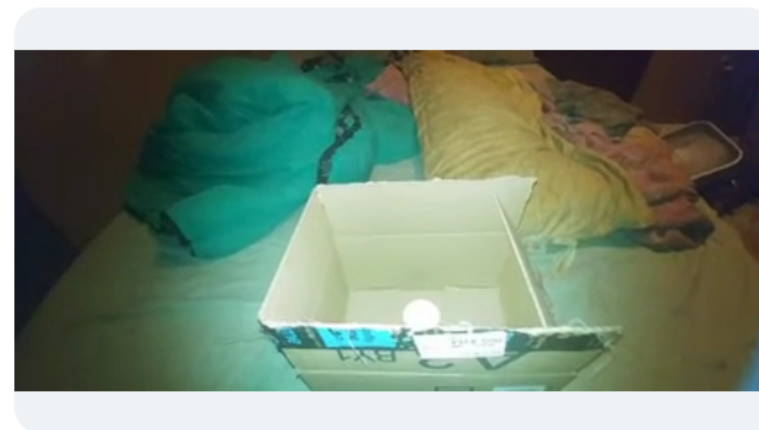
Predict a future state — in feature space

Now



“Put the jar into the box”

Later



Predicted future features

Match

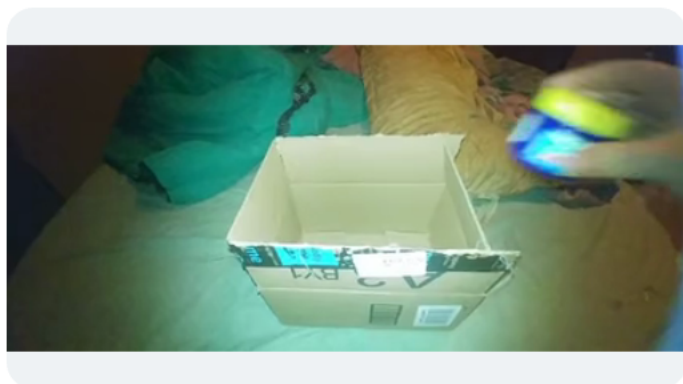


Encoded future target

Like anticipating “jar inside box” — without drawing the next frame.

The tokens describe an intended transition

Current observation



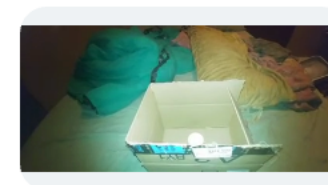
Put the jar into the box

VLM
Qwen3-VL-2B



Latent action tokens

An intended transition

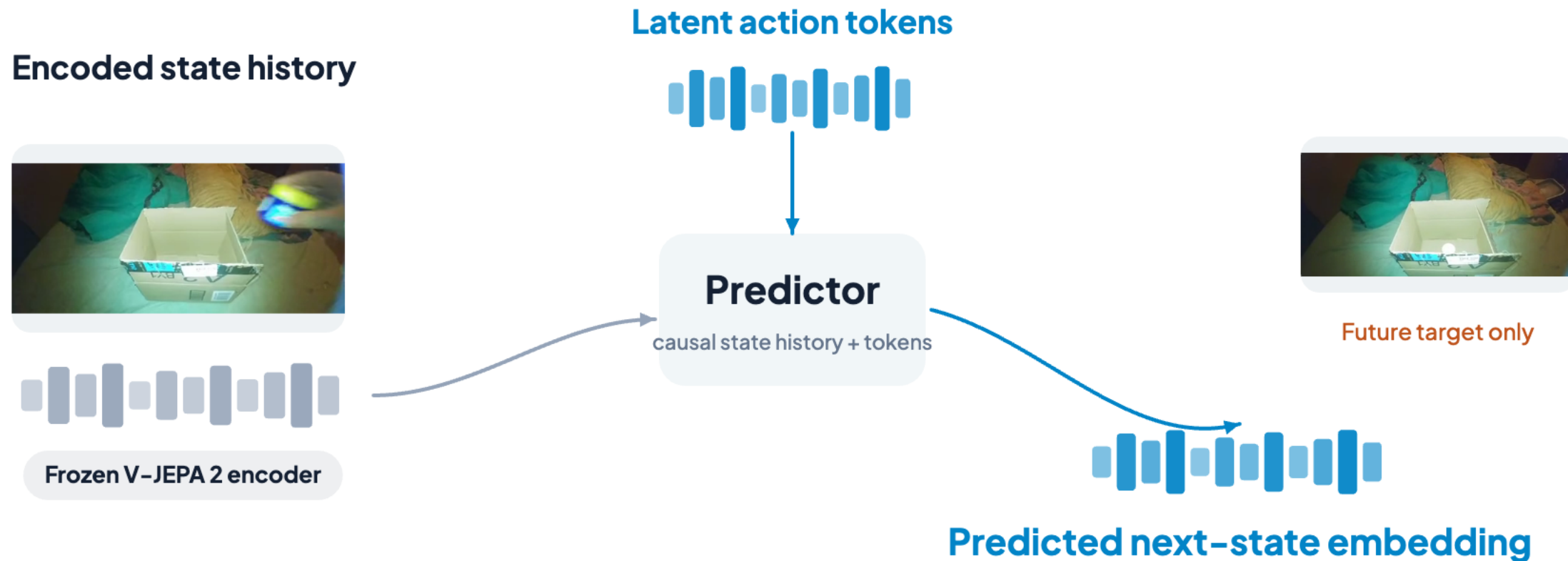


Supervision only

These tokens condition both prediction and control.

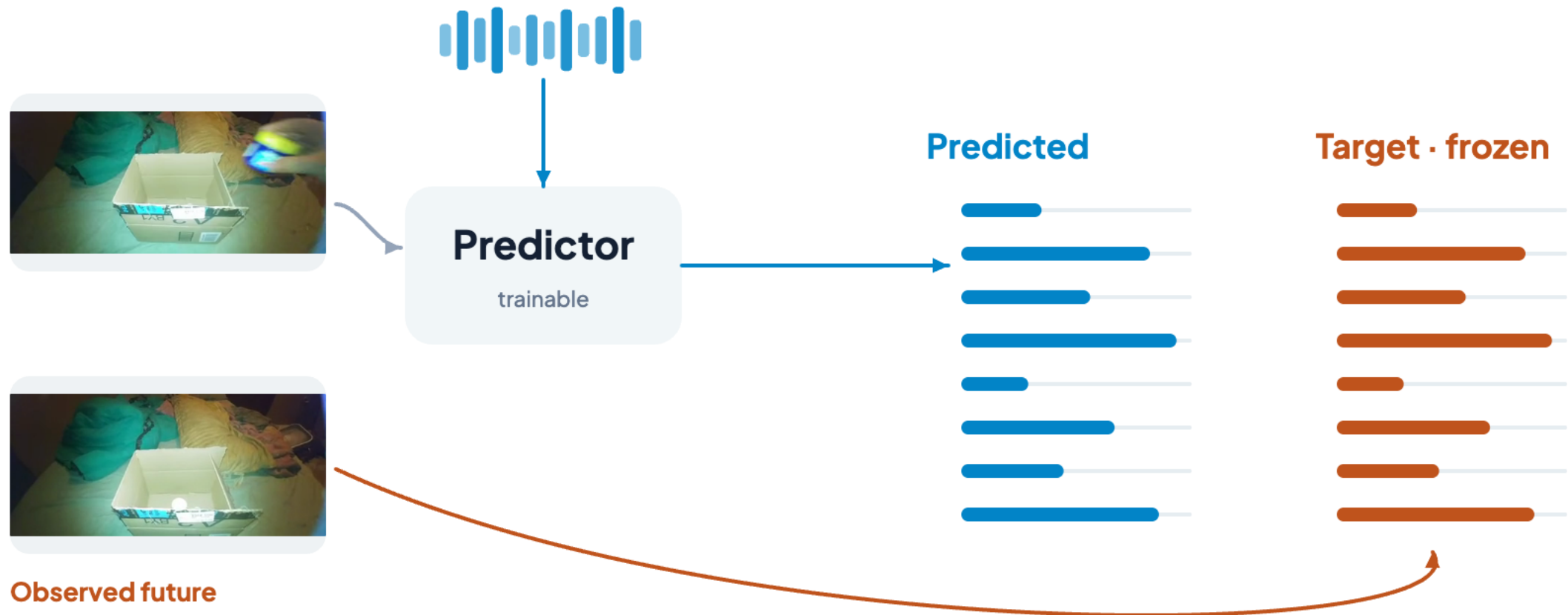
They are neither pixels nor executable motor commands.

The predictor outputs the next state's features



The output describes the next state; the tokens describe the transition.
No image decoder is needed for this supervision objective.

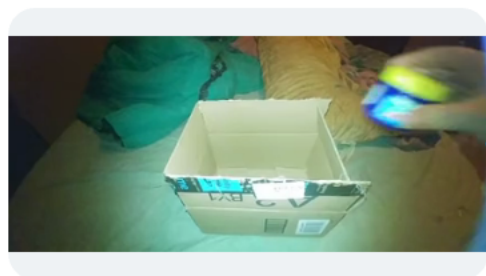
The target stays fixed. The prediction learns.



Training makes noisy corrections, not a literal motion between states.

The policy never receives its future target

Initial observation



Policy → latent tokens



Predictor

causal history

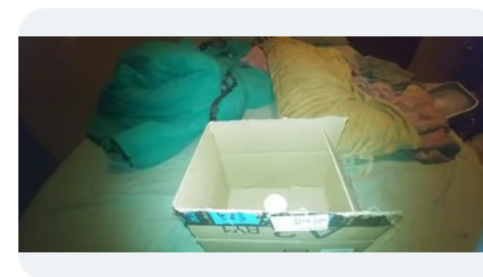
Prediction



Training target only

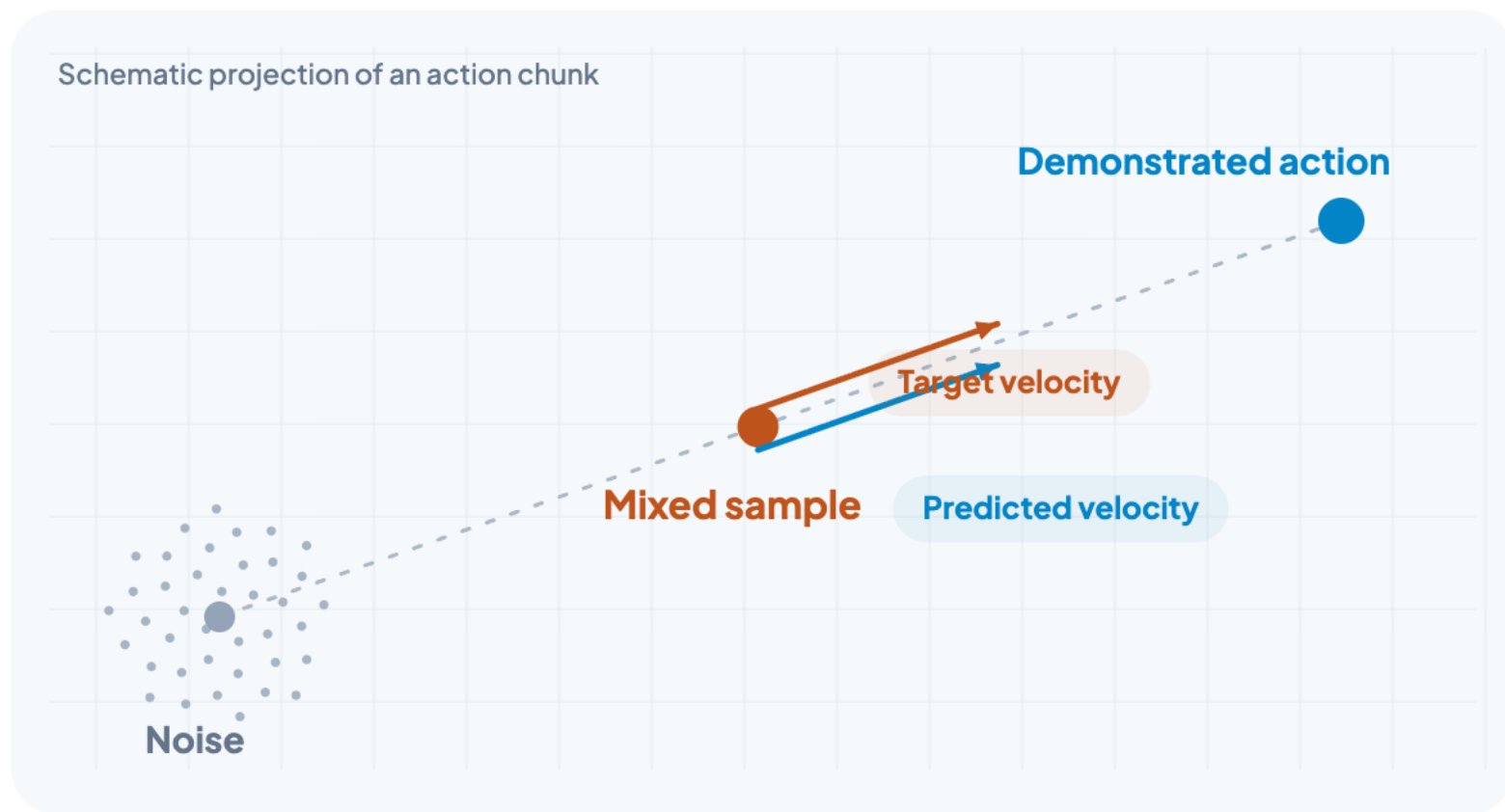
Future frames stay below this boundary.

Compare only



At deployment: observation + instruction → tokens → action head.

Teach the field with noise–demonstration pairs



Latent tokens



Velocity field

conditioned on tokens + time

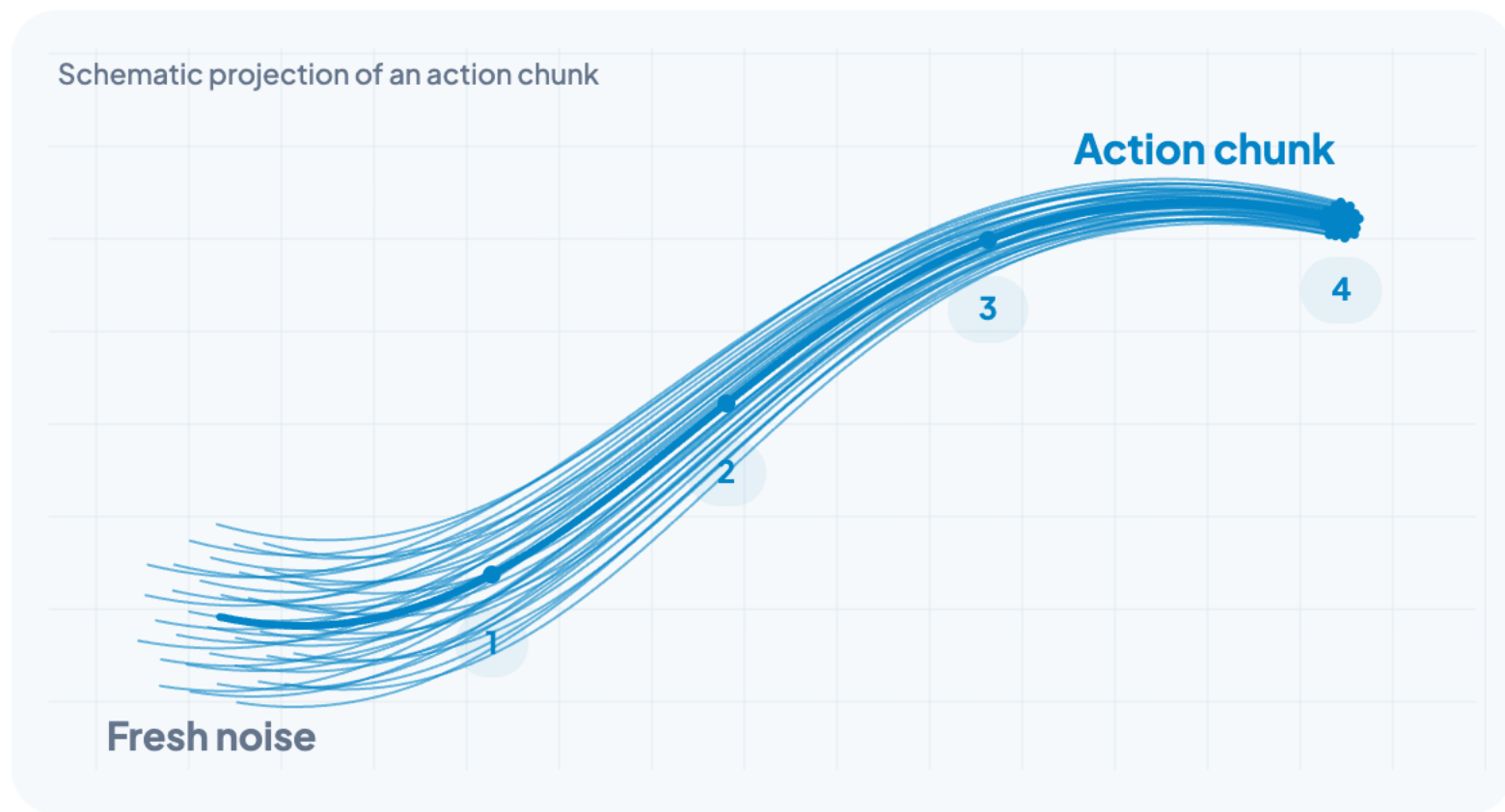
Sample a pair

Choose a mixing time

Match the velocity

Robot demonstrations teach which directions lead to valid control.

Now follow the field from a fresh noise sample



Latent tokens



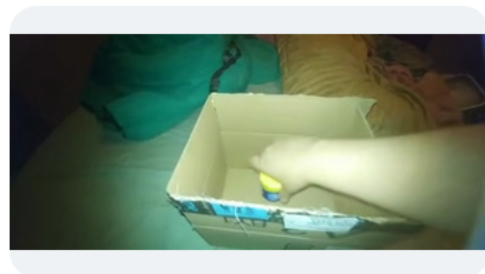
Velocity field

conditioned on tokens + time

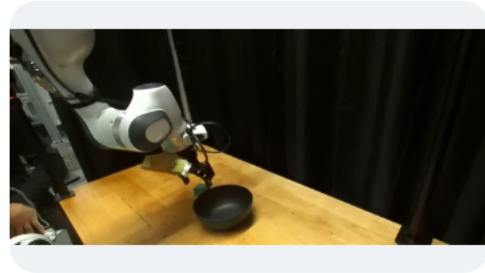
Start from fresh noise
Re-evaluate the field
After each small update

The action head generates controls; it does not decode a predicted image.

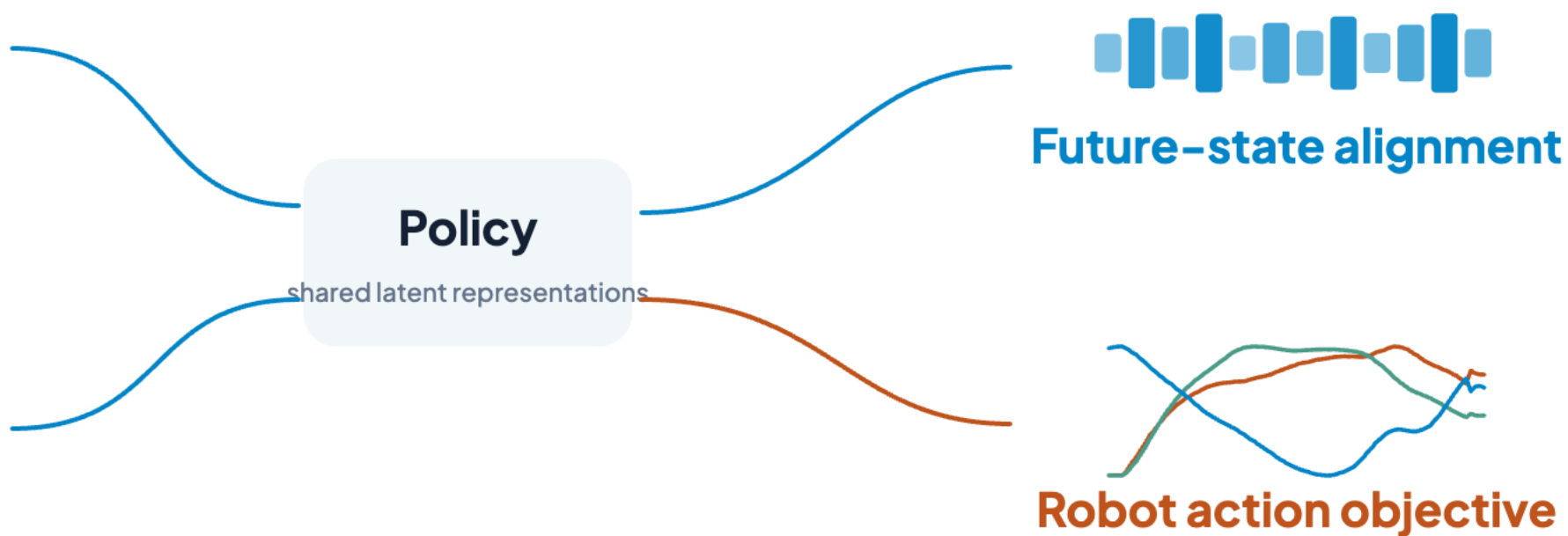
Video teaches prediction. Robot data also teaches control.



Human video



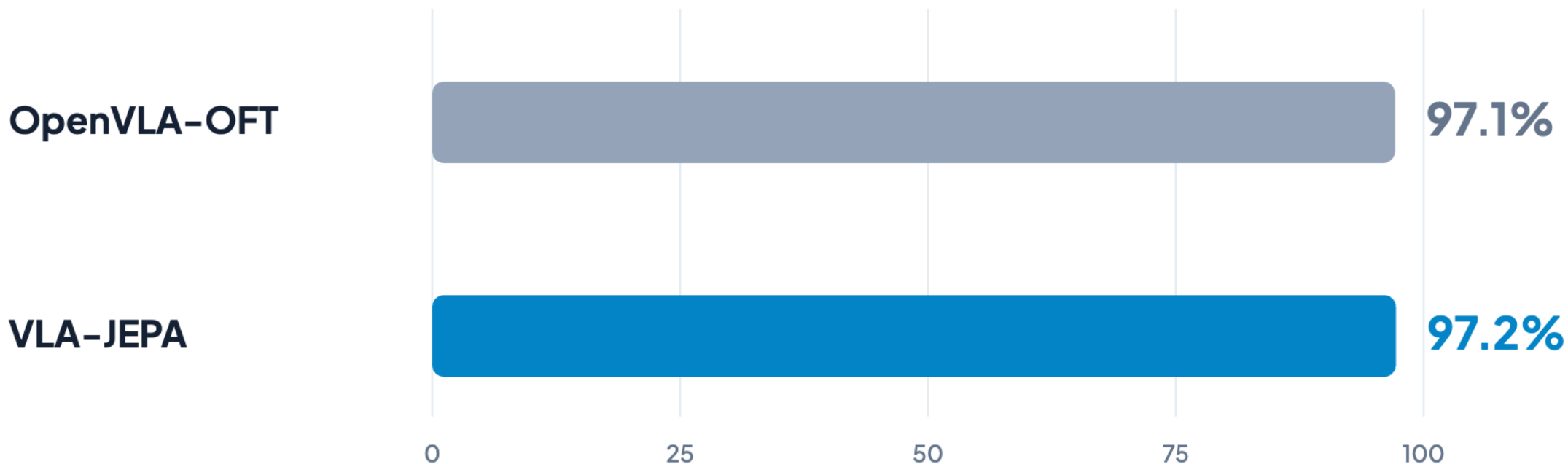
Robot demonstration



Human data shapes the representation; labelled robot data anchors it to control.

Standard LIBERO: a near tie

Success rate



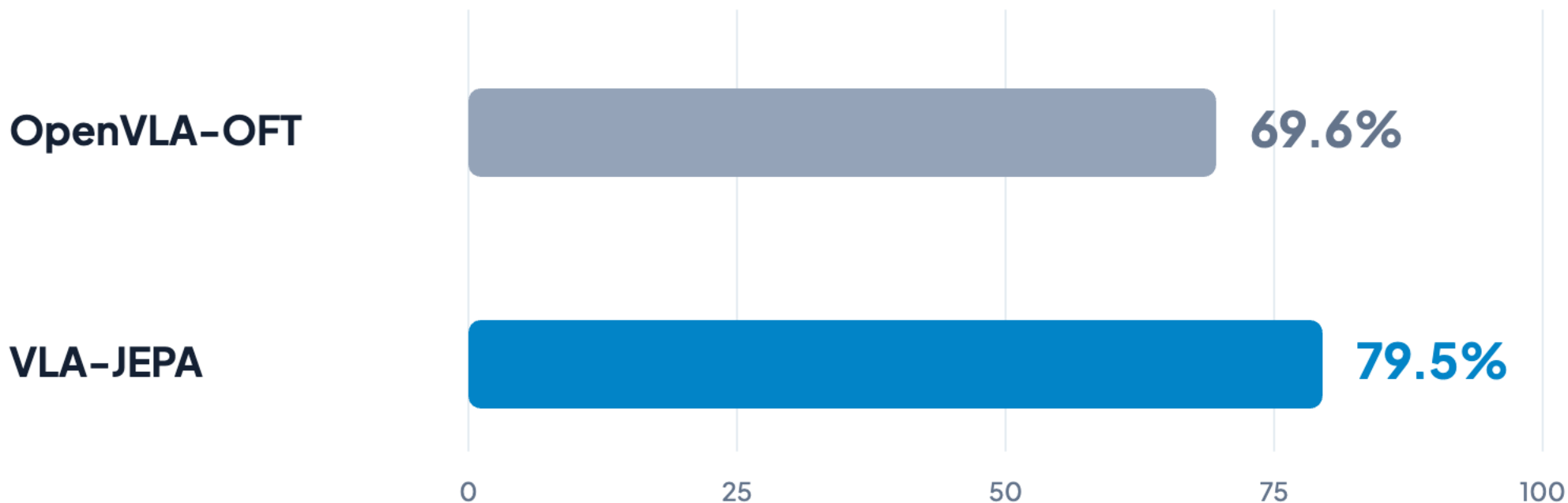
+0.1 percentage points

A near tie on standard tasks does not establish a meaningful advantage.

Paper Table 1 · Success rate (%) · No reported confidence intervals

LIBERO-Plus: stronger under perturbations

Success rate

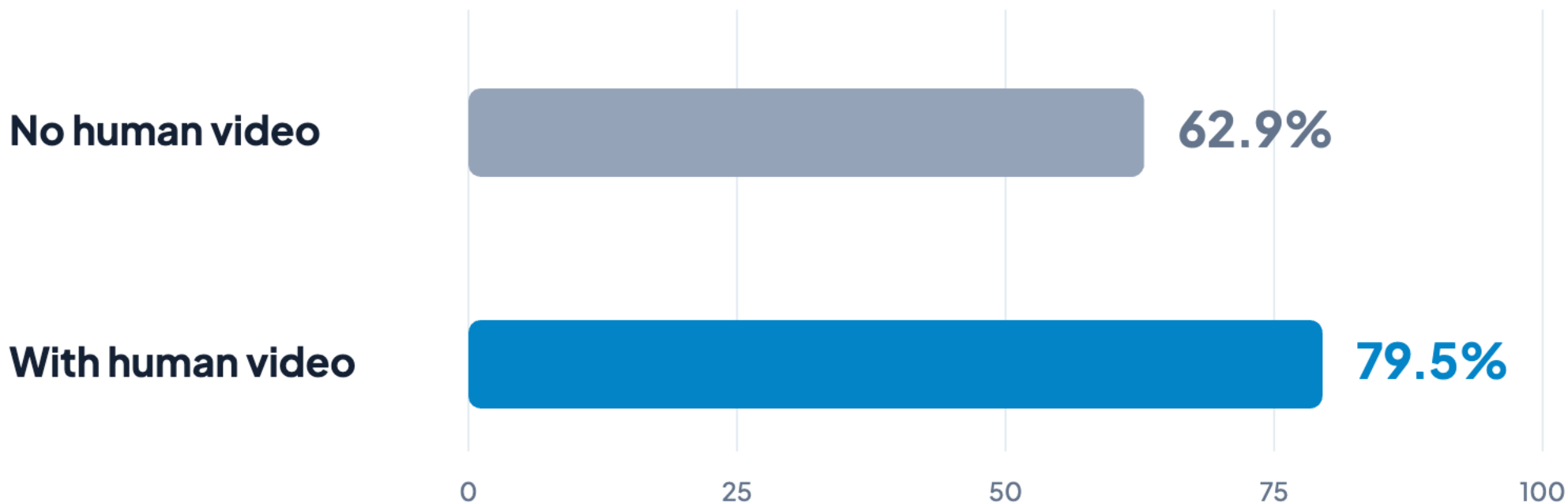


+9.9 percentage points

The stronger claim is resilience under changed conditions.

Human video improves LIBERO-Plus robustness

Success rate

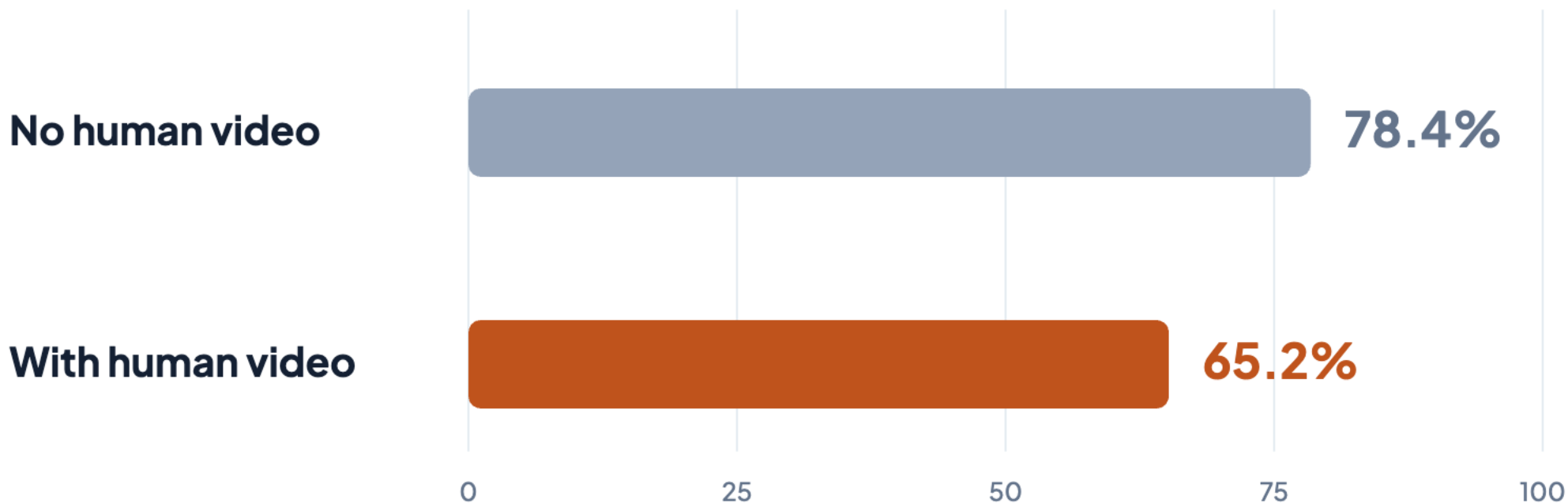


+16.6 percentage points

Compare the same method, with and without the extra video data.

Human video does not improve every task

Success rate



-13.2 percentage points · SimplerEnv Google

The benefit depends on the benchmark; human video is not uniformly helpful.

Can we lower the cost and keep useful control?

01

Reduced precision

Compare against the same baseline

Measure memory, latency and success

02

Smaller backbone

Explore SmolVLM integration

Keep prediction and action components

03

Work so far

Implementation and evaluation setup

Validate before drawing conclusions

Matched baseline → resource use + task success

The trade-off is a question to test, not an improvement to promise.

Strong averages leave deployment questions open

SENSOR NOISE

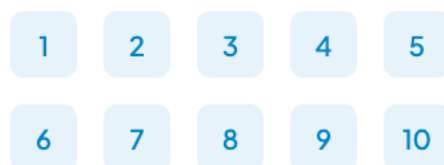
66.3%

VLA-JEPA

79.0% · pi zero

A robustness counterexample

REAL-WORLD EVIDENCE



10 trials per task

Wrong-object selections

TRAINING → DEPLOYMENT



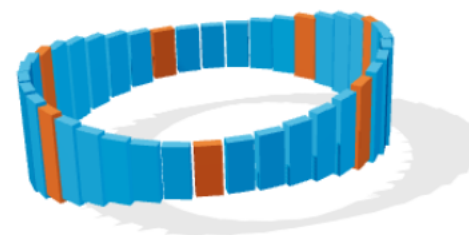
8 A100 GPUs · paper training

Proposed study: single-GPU efficiency

Headline averages do not establish deployment reliability.

No reported confidence intervals on these headline comparisons.

Predict states. Learn control.



Strongest evidence

Robustness under perturbations

Our open question: lower resource cost
while keeping useful control?

A smaller backbone is a hypothesis to evaluate



Keep the video encoder, world predictor and action head.
Check integration and matched evaluation before concluding.

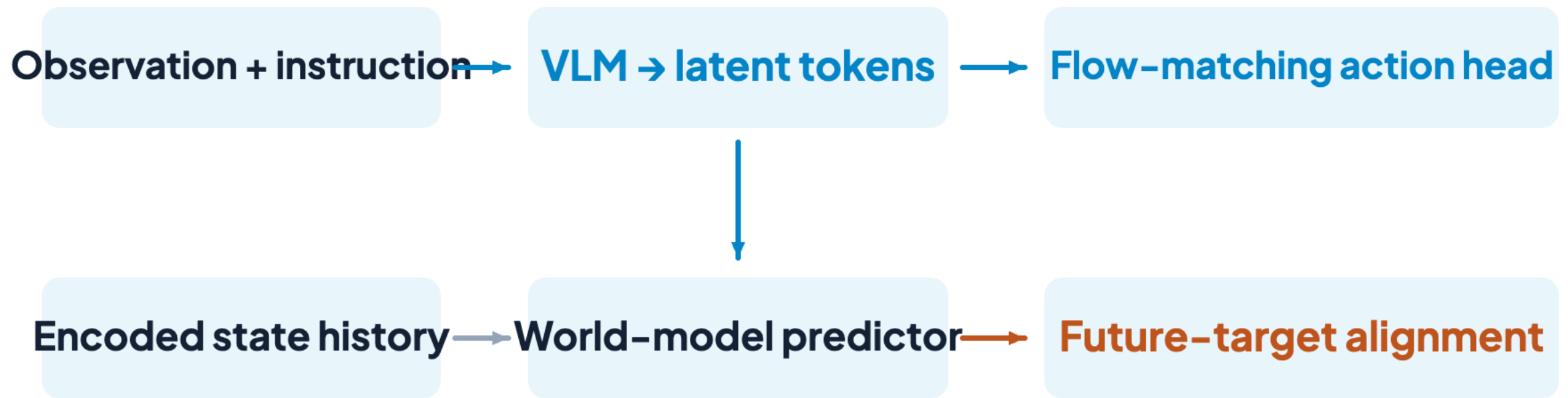
Architecture search comes after a reliable comparison



An optional direction, with an open outcome

Resource savings would need evidence of useful robot control.

The whole method, after the components are familiar



Video supervises prediction. Robot demonstrations also supervise actions.

The equations behind the visual training pair

Mix noise and a demonstrated action chunk

$$a_{\tau} = (1 - \tau)\epsilon + \tau a^*$$

Regress velocity onto the displacement of the pair

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\left\| v_{\theta}(a_{\tau}, \tau | z_a) - (a^* - \epsilon) \right\|_2^2 \right]$$

Mixing time describes generation progress, not physical robot time.